THE MODULE GENERATED BY A SEQUENCE WITH KOSZUL RELATIONS SURJECTS ONTO THE IDEAL WITH KERNEL ANNIHILATED BY IT

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- The module presented by generators e_i and the relations f_i e_j = f_j e_i. the statement itself, source lines 24394-24397. Supplies the module M and the relations that the computation uses.
- The Koszul complex of a sequence and the identification of its first homology. recalled in the first sentence of the proof, source line 24405, referring to the theory of the same chapter. Explains why the presentation is the truncation of the Koszul complex and why the kernel is what it is.
- An element of the kernel of M to I can be written as a finite combination of the generators and satisfies the corresponding relation in the ideal. used at proof lines 24406-24409. Is the form in which the computation of f_j m is carried out.

Source reference. Original lemma 05ED: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6986–6986. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Lemma 0.2. *Let R be a ring. Let $I = (f_1, \ldots, f_n)$ be a finitely generated ideal of R. Let M be the R-module generated by elements $e_1, \ldots, e_n$ subject to the relations*

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$f_i e_j - f_j e_i = 0$. There exists a short exact sequence

$$0 \to K \to M \to I \to 0$$

such that K is annihilated by I.

Proof. This is just a truncation of the Koszul complex. The map $M \to I$ is determined by the rule $e_i \mapsto f_i$. If $m = \sum a_i e_i$ is in the kernel of $M \to I$, i.e., $\sum a_i f_i = 0$, then $f_j m = \sum f_j a_i e_i = (\sum f_i a_i) e_j = 0$. □